International Journal of Disaster Risk Reduction 128 (2025) 105725

Contents lists available at ScienceDirect

International Journal of Disaster Risk Reduction

journal homepage: www.elsevier.com/locate/ijdrr

Evaluation of psychology-based training for improving interoperability in the emergency services: A comparison of online and in-person delivery methods

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ARTICLE INFO

Keywords:
Interoperability
Emergency response
Training
Social identity
Inter-organizational working
Multiteam systems

ABSTRACT

Recent research highlights the importance of incorporating psychological perspectives to improve interoperability in multi-agency emergency response in order to address persistent challenges and prevent future issues. To address this, we developed a psychology-based training programme grounded in the Social Identity Approach. Its effectiveness and optimal delivery method were evaluated with 65 emergency responders from UK Police ($N = 8$), Fire and Rescue ($N = 12$), and Ambulance ($N = 45$) Services. Participants completed the training either online ($N = 28$) or in-person ($N = 37$), with follow-up interviews conducted with seven of the online participants. In terms of participant satisfaction, the training was positively received and recommended by participants. They valued the psychological elements but stressed the need for accessible presentation. Interviewees preferred in-person training, but survey data showed no difference in participant satisfaction between delivery methods. In terms of knowledge gain, both delivery methods increased confidence in multi-agency teamwork, though in-person training better enhanced knowledge of specific collaborative actions. Participants highlighted the importance of understanding responders' motivations, especially regarding mandatory training. This evaluation offers valuable insights into the design and delivery of effective emergency service training and demonstrates how integrating psychological theory can support better interoperability in multi-agency contexts. Practical implications are discussed.

1. Introduction

For over a decade, improving interoperability (i.e., the way responders from different organisations work together) among UK emergency services has been a priority. To aid this, the Joint Emergency Services Interoperability Programme (JESIP) was introduced in 2012, offering standardised principles for joint working [1]. These principles include clear communication free from technical jargon and co-locating at a single, safe location. Despite these efforts, interoperability problems persist, hindering effective collaboration [2].

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https://doi.org/10.1016/j.ijdrr.2025.105725
Received 24 July 2024; Received in revised form 22 July 2025; Accepted 22 July 2025
Available online 23 July 2025

A review of 32 major incidents in the UK from 1986 to 2010, including the 1989 Hillsborough Stadium Disaster and the 2005 London bombings, identified persistent interoperability problems, such as poor communication, unclear roles, and a lack of leadership [3]. Despite the introduction of JESIP, these issues continued in the 2017 Manchester Arena attack [4] and the Grenfell Tower fire [5]. For example, conflicting communication during the Manchester attack prevented a multi-agency meeting point, leading Saunders [6] to note that "JESIP still failed on 22nd May 2017" (p.135), with responders seemingly working in silos.

These challenges are not unique to the UK's emergency response either. Internationally, multi-agency responses face similar issues. For example, Comfort and Haase [7] reviewed the 2005 Hurricane Katrina response in America, finding limited collaboration between organisations despite guidelines to facilitate joint efforts, partly due to poor communication alignment among emergency services. The 2010 Haiti earthquake response also faced coordination issues, with excessive information sharing making identifying relevant information difficult [8]. Similar problems have been observed in Norway [9], Australia [10], and the Netherlands [11], demonstrating the prevalence of interoperability issues globally. Therefore, there is a clear need to better understand why interoperability challenges occur in order to inform multi-agency training and guidance, and ultimately improve practice.

1.1. Understanding interoperability

Interoperability involves responders from different organisations forming a multiteam system, where interdependent sub-teams work together as a multiteam system towards a collective goal [12,13]. Each sub-team has different goals, such as neutralising threats (Police) or treating casualties (Ambulance), together contributing to the shared goal of saving lives and reducing harm [14]. However, responders often fail to collaborate as one team, instead working in silos, as seen in the Manchester attack [4].

To improve our understanding of multi-agency collaboration and provide insights into why interoperability challenges might persist, Davidson and colleagues recently developed a social identity perspective on interoperability [15–18]. This perspective builds on social identity research in other multiteam system domains, such as military teams [19]. The Social Identity Approach is a key framework in social psychology for understanding within- and between-group behaviour. When people perceive themselves to be in the same group, this is the basis for shared social identity, in other words, a shared sense of 'us-ness' [20,21]. Here, rather than being defined as 'I' and 'me', shared identity defines the self as 'we' and 'us' in ways that psychologically connect people. Importantly to the problem of interoperability, research suggests that a shared identity is linked to increased motivation of group members to work together as well as improved group performance [22]. It follows too that shared identity can also influence and facilitate several interpersonal factors, such as communication, information sharing, and trust [23]. Theoretically then, if responders see themselves as part of a unified 'we' as emergency responders, rather than sub-team 'we's' as Police, Fire and Rescue Service (FRS), or Ambulance responders, they are more likely to coordinate and cooperate due to a stronger sense of interconnection, shared understanding, common norms, and increased trust [15–17,22,24].

With this in mind, Davidson and colleagues explored the role of shared identity in multi-agency emergency response through two interview studies with responders involved in the COVID-19 response in the UK [15,16]. They found that structural barriers, such as unequal building access and different shift patterns, hindered the development of a shared identity between responders from different organisations. Despite these barriers, both studies showed that shared identity could still arise, facilitated by a common purpose from the shared threat of COVID-19, sharing difficult experiences, and leaders emphasising shared goals. Advancing on this, discussion-based simulated exercises found that shared identity was associated with improved interoperability through increased motivation, confidence, trust, and respect among responders [17]. A recent systematic review supports the research of Davidson et al., identifying identity, trust, and cohesive goal setting as key psychological factors for effective interoperability (Power et al., 2023).

1.2. Translating psychological principles of interoperability into practice

The evidence above highlights the importance of understanding the psychological aspects of interoperability. However, there is little evidence on how to apply these psychological principles in practice, reflecting a well-documented research-practice gap in emergency response [25,26]. Evidence of this gap in relation to interoperability can be seen in the persistent issues and repeated recommendations for improved joint working, identified following a review of 52 local response debrief reports across England, Scotland, and Wales [2,27,28]. This suggests that identified lessons are often not translated into learning, highlighting the need for solution-focussed approaches to facilitate knowledge transfer and address this ongoing challenge.

A successful example of knowledge transfer in emergency response is in the field of mass decontamination for chemical, biological, radiological, and nuclear (CBRN) incidents. Findings from research by Carter and colleagues [29–31] on the psychosocial aspects of incidents requiring mass decontamination were developed into a training module that is now incorporated into UK FRS training for the management of CBRN incidents [32]. This demonstrates the impact that research can have on practice when presented in an accessible way, helping to minimise the research-practice gap.

However, in terms of how training is delivered, previous research offers mixed evidence on whether online training is as effective as in-person training in facilitating knowledge transfer. Some studies suggest that in-person training is more effective at increasing knowledge (e.g., Ref. [33]). They attribute this to factors such as participant preparation, the physical environment (e.g., at home versus at work), and instructional materials not being properly adapted for online delivery. Conversely, other studies, such as Gallegos et al. [34], found no significant differences between the two delivery methods. Interestingly, Mallonee et al. [35] found no difference in knowledge gains between online and in-person evidence-based psychotherapy training. However, participants in the online training reported lower satisfaction compared to those in in-person training. The authors speculated that this might be due to participants being more accustomed to in-person training. Despite this evidence, there is still limited research examining how different training delivery methods affect emergency responders. Whilst previous research has shown that responders tend to prefer in-person training [36], the effectiveness of online training in this context has yet to be fully explored. With this in mind, the current study compares online and in-person delivery methods to assess the suitability of each for delivering psychology-based interoperability training in the emergency services to provide insights into this knowledge gap.

1.3. Current UK emergency services interoperability training

JESIP models and principles set the standard for interoperability in the UK emergency services. The Joint Doctrine [1] provides a standardised approach and offers training opportunities, including a one-day multi-agency training course and various awareness products (e.g., e-learning, films, mobile applications). These resources cover topics such as the need for interoperability, the impacts of declaring a major incident, and the joint working principles, focussing on practical actions like clear and jargon-free communication. Power et al. [37] argued however that psychological processes are as important as practical actions. Yet, despite this recognition, psychological processes have not yet been embedded into interoperability training. This paper aims to address this gap by first detailing the development of a novel psychology-based interoperability training, and then present findings from its initial evaluation.

1.4. Aims and objectives

Based on key psychological principles of interoperability identified within recent research ([15–17]; see Ref. [18] for discussion), we developed psychology-based training to improve interoperability and contribute to addressing the research-practice gap within the emergency services [25,26]. The purpose of the current paper is to outline the development of this training and evaluate it. Specifically, we explored whether responders found the psychological elements of the training useful for improving their knowledge of how to collaborate with other responders. In addition, we also explored whether there were any differences between online and in-person deliveries of the training in participants' engagement with the training and their perceived competence of working interoperably.

2. Method

2.1. Training development

The development of the training consisted of two main components: (i) pedagogic considerations and (ii) content selection, which this section addresses in turn. A co-production approach to training development was used with eight emergency responders from the three blue-light services. Draft versions were shared with them, and their feedback was used to refine and finalise the materials. Details of their input are provided in Appendix 1.

2.1.1. Pedagogic considerations

The training was grounded in the recent application of the Social Identity Approach to interoperability [18] and drew on insights from established theory-based training initiatives. These include the Mass Decontamination Instructors Course [32], the 5R Leadership Development Programme [38], and Groups for Health (G4H; [39,40]). Each of these programmes exemplifies a 'theory-into-action' approach, translating theoretical principles into practical strategies to build shared identity and enhance team performance.

A large body of evidence attests to the accessibility and efficacy of social identity-based programmes. For example, the 5R Leadership Development Programme was evaluated over a two-year period with leaders of the British national paralympic football team [41]. The findings showed significant increases in social identification among staff, as well as notable rises in the number of training hours athletes completed outside of official training camps. This suggests that the 5R programme benefits not only the leaders themselves, but also improves the performance of the teams they guide (cf. [23,39]). Further evidence highlights the positive impact of the G4H programme on individual well-being and group connectedness [39,40,42], with some studies demonstrating sustained benefits lasting up to two years post-intervention [41,43]. Taken together, this body of research provides a strong foundation for the development of our own social identity-inspired training programme.

For the structure of the training, we chose a single-session approach, like the Mass Decontamination Instructors Course, in light of the time pressures and operational commitments that responders often face [44]. To enable scalability, we developed a version of the training that could also be delivered online. Given the time pressures emergency responders are often under, we wanted to explore the effectiveness of different delivery methods and establish whether online training could be an effective way of delivering the training.

2.1.2. Content selection

The training content was designed around three key learning outcomes, aligned with the overarching goal of using psychology to improve interoperability. It was evidence-based and grounded in theory drawn from several domains, including emergency response, group behaviour, and interoperability.

The first learning outcome was to increase understanding of the challenges in the implementation of interoperability. This section explored real-world issues in multi-agency collaboration, drawing on evidence from past incidents. Key examples included the Pollock review (2013), the 2017 Manchester Arena attack [4], and an overview of the JESIP principles [1].

The second learning outcome was to increase understanding of how group psychology can be used to improve interoperability. The Social Identity Approach was introduced here, which explains how group psychology can improve coordination. A social identity mapping task [45,46] was used in this section, where participants reflected on group dynamics in a recent multi-agency incident they attended. The relevance of the Social Identity Approach was reinforced through examples from other emergency response settings, such as emergency decontamination [31] and riot diffusion [47], as well as broader organisational contexts [48].

The final learning outcome was to improve understanding of the specific actions that responders can take to improve interoperability in their work. Here, three practical, theory-driven points were emphasised. The first of these was to 'build relationships with other services.' Based on the concept of identity 'impresarioship', this point underscored the role of shared experiences in creating a shared identity and improving coordination [15,38,49]. The second key point was to 'use your shared frame of reference.' Drawing from the concept of identity enactment, this point explained how shared norms and values guide behaviour within a group [15,16,50,51]. The final point was 'make your common purpose clear.' Based on the concept of collective agency, this focused on how shared goals and mutual understanding of roles can help facilitate effective collaboration [15,16,47].

2.2. Delivery methods

The training was designed to be delivered either online or in-person, with both methods featuring identical content and structure. For the online delivery, the first author recorded a 30-min video presenting the training. The video included 'stop and think' sections, where participants were asked to pause and reflect on specific topics (e.g., the different teams typically involved in a multi-agency response).

For the in-person delivery, the first author presented the training directly to responders, with the session lasting approximately 45 min. Although the content was the same, the 'stop and think' sections in the in-person format included facilitated discussions, allowing participants to share their thoughts with each other and provide feedback to the researcher and the larger group. Additionally, participants could put questions to the researcher or their peers at any time during the session.

2.3. Evaluation design

The evaluation used a mixed methods design, with both quantitative and qualitative data being collected. Quantitative data was collected through a questionnaire that participants completed before and after receiving the training. Qualitative data was collected through semi-structured interviews with a subset of participants after completing the training.

2.4. Participants

Sixty-five participants from the Police ($N = 8$), FRS ($N = 12$), and Ambulance Service ($N = 45$) completed the training evaluation, either online ($N = 28$) or in-person ($N = 37$). For the online delivery, participants were recruited through opportunity sampling via social media, emergency service contacts, and word of mouth. For the in-person delivery, the first author attended a pre-organised commander development day. All in-person participants were from one Ambulance Service in the South of England. Online

Table 1
Participant information provided in frequency (percentage).

Organisation	
Police	8 (12.31 %)
FRS	12 (18.46 %)
Ambulance	45 (69.23 %)
Command level	
Operational	10 (15.38 %)
Tactical	25 (38.46 %)
Strategic	6 (9.23 %)
Operational and Tactical	10 (15.38 %)
Operational, Tactical, and Strategic	5 (7.69 %)
Other	9 (13.84 %)
Sex	
Male	54 (83.08 %)
Female	7 (10.77 %)
Not stated	4 (6.15 %)
Years in service	
0–5	2 (3.08 %)
6–10	7 (10.77 %)
11–15	12 (18.46 %)
16–20	13 (20.00 %)
21–25	15 (23.08 %)
26–30	8 (12.31 %)
31+	4 (6.15 %)
Not stated	4 (6.15 %)
Familiarity with JESIP	
Moderately familiar	1 (1.54 %)
Very familiar	22 (33.85 %)
Extremely familiar	38 (58.46 %)
Not stated	4 (6.15 %)

participants were from Police, FRS, and Ambulance Services across the UK. Eligible participants were current or former operational, tactical, or strategic commanders in the UK emergency services. Participants had a range of experience, with most reporting being ‘very’ or ‘extremely’ familiar with JESIP (see Table 1 for participant information). Prior to participation, participants confirmed they had not previously viewed the training video. At the end of the online delivery, participants indicated if they were willing to complete a short follow-up interview, of which seven agreed (Police, $N = 2$; FRS, $N = 3$; Ambulance, $N = 2$).

2.5. Materials

2.5.1. Pre-training questionnaire

This questionnaire included four questions about participants’ perceived competence working interoperably (e.g., “I think it is important to work with responders from other services”), rated on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). See Supplementary Materials 1 for the full questionnaire.

2.5.2. Post-training questionnaire

This questionnaire included the same four questions as the pre-exercise one, plus nine additional questions about participants’ engagement with the training (e.g., “I think that this training has improved my knowledge on interoperability”). Participants rated each statement on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). See Supplementary Materials 2 for the full questionnaire.

2.5.3. Interview guide

The interview guide included similar questions to those in the post-training questionnaire (see Supplementary Materials 3 for the interview guide).

2.6. Procedure

Online participants completed the evaluation via Qualtrics from 25th February to 6th April 2023. They received an information sheet explaining the evaluation and they provided consent before completing the pre-training questionnaire. They then watched the training video before completing the post-training questionnaire. At the end, participants indicated if they wanted to take part in a follow-up interview. These interviews were conducted by the first author on Microsoft Teams from 23 March to 12 April 2023 and lasted an average of 37 min.

In-person participants took part on 26th May 2023 during an Ambulance Commanders Development Day. They completed the pre-training questionnaire then received a 45-min training session delivered by the first author. After the training, participants completed the post-training questionnaire.

2.7. Analysis

2.7.1. Questionnaire data

The quantitative questionnaire data was analysed using SPSS 28.0.1. One-sample *t*-tests were conducted to explore participants’ engagement with the training by comparing participants’ average score to the midpoint value of 3 on the Likert scale. Independent samples *t*-tests were conducted to compare participants’ engagement between online and in-person deliveries. Paired samples *t*-tests were conducted for comparing pre- and post-training scores for perceived competence in both delivery methods. Spearman’s rank-order correlations were conducted to examine the relationship between responders’ years in service and their engagement with the training.

A post-hoc effect-size sensitivity analysis was conducted to explore if we had enough power to detect effects [52]. Sensitivity analyses were conducted for each statistical test using G*Power [53]. Results show the analysis was partially underpowered (see Appendix 2).

2.7.2. Interview data

Interview data were analysed using semi-deductive thematic analysis [54]. The first author transcribed and read the interview transcripts, noting points relevant to the evaluation objectives. These notes were used to develop codes and relevant data extracts were categorised accordingly. After discussing the codes with the research team, themes were identified and defined. The first author then re-read the transcripts, organising data into these themes. The final analysis was reviewed and agreed upon with the research team. Four themes were identified: inclusion of psychological theory, delivery method, importance of collaboration, and motivation for training.

3. Results

3.1. Questionnaire data

For participants’ engagement with the training, one-sample *t*-tests revealed that overall, participants rated the training positively – responses to questions asking about participants ability to follow the information in the training, whether participants understood the training, whether they found the training helpful, and whether they would recommend it to others were significantly above the midpoint value of 3 (see Table 2). No significant differences were found in participants' engagement with the training between the online and in-person delivery (see Table 3).

For those who completed the online training, paired samples *t*-tests revealed a significant increase in participants' perception of their ability to work effectively in a multi-agency team as well as participants' confidence to be able to effectively work in a multi-agency team, after taking part in the training compared to before. No significant differences were found in participants' knowledge of actions they could take to facilitate an effective multi-agency response and their perception of the importance of working with responders from other services before completing the training compared to after (see Table 4).

For the in-person delivery, paired-samples *t*-tests revealed significant differences in participants' confidence to be able to effectively work in a multi-agency team, as well as their knowledge of actions they could take to facilitate an effective multi-agency response, with participants reporting significantly higher agreement scores after taking part in the training compared to before. No significant differences were found in participants' perceived ability to work effectively in a multi-agency team, or in their perceptions of the importance of working with responders from other services before completing the training compared to after (see Table 4). In addition, no significant differences were found in the amount of change of these variables between online and in-person groups (see Table 5).

Spearman's rank-order correlations were conducted to examine the relationship between responders' years in service and their engagement with the training. A significant positive correlation was found between years in service and the belief that the training was helpful, $r(65) = .26$, $p < .05$, indicating those with more years in service were more likely to find the training helpful than those with less years in service. No other correlations reached statistical significance, although small, non-significant positive correlations were observed between years in service and liking the training ($r = .23$), perceived importance of the training ($r = .18$), recommending the training to others ($r = .21$), improved knowledge ($r = .10$), improved own skills ($r = .05$), and improving others' ability ($r = .10$).

3.2. Interview data

This section presents the results from interviews with a subset of participants who completed the online training delivery. Results are separated into four themes: inclusion of psychological theory, delivery method, importance of collaboration, and motivation for training (see Table 6).

3.2.1. Inclusion of psychological theory

Nearly all participants expressed appreciation for the psychology and science components of the training. They valued the evidence and reasons provided for following JESIP, noting that previous training lacked this evidence base, for example "every programme I've been on tells us the practical, this is what you should do […] but no one has ever really explained the science […] but now I understand that there is science behind it" (P5).

However, three participants raised concerns about certain aspects of the psychological theory presented. One felt the training was more academic than others they had experienced and stressed the need to make it accessible to all participants by simplifying the language and the content. Another participant echoed this, suggesting that less academic language would prevent creating "unhelpful barriers" when trying to engage responders with the training.

Despite these concerns, two of the participants who critiqued the content also mentioned they would have liked to see more psychology included, as it made the training unique. Another participant agreed, stating that the psychology elements addressed a "gap in the market" for interoperability training and expressed a desire for even more psychological content.

3.2.2. Delivery method

Most participants said they preferred in-person training over online delivery. However, many acknowledged the value of online training when in-person sessions are not feasible. They recognised that different people have different learning styles and that a "one-size-fits-all" approach is not ideal. These participants stressed the importance of tailoring online training to its specific format, considering that participants cannot ask questions directly.

Some participants highlighted the advantage of online training in reaching a larger, more geographically dispersed audience quickly. One participant noted, "for what you have, the massive audience that you've got to try and reach, and the geographical spread of it, I think [online delivery] is a really good solution, and it's a practical solution" (P5).

Two participants suggested the training should ideally be offered both in-person and online to reinforce learning. They felt that in-person training would allow for follow-up on the information provided, while online training would offer flexibility in when and where

Table 2
Combined online and in-person one sample *t*-test results for participants' engagement with the training.

Item	Mean (standard deviation)	*t*-statistic	*p*	Standard error
Found it helpful	4.28 (.70)	14.79	<.01	.70
Liked the training	4.25 (.71)	14.20	<.01	.71
Improved knowledge	4.02 (.72)	11.40	<.01	.72
Will improve skills	4.03 (.71)	11.76	<.01	.71
Will improve others' ability	4.18 (.58)	16.37	<.01	.58
Think the training is important	4.28 (.65)	15.85	<.01	.65
Recommend the training to others	4.34 (.62)	17.42	<.01	.62

Table 3
Independent sample *t*-test results comparing online and in-person training for participants' engagement with the training.

Item	Mean (standard deviation)		*t*-statistic	*p*	Standard error
	Online	In-person			
Found it helpful	4.36 (.68)	4.22 (.71)	−.81	.42	.70
Liked the training	4.36 (.68)	4.16 (.73)	−1.10	.28	.71
Improved knowledge	4.04 (.74)	4.00 (.71)	−.20	.84	.72
Will improve skills	3.96 (.79)	4.08 (.64)	.66	.51	.71
Will improve other's ability	4.21 (.69)	4.16 (.50)	−.35	.72	.59
Think the training is important	4.29 (.66)	4.27 (.65)	−.09	.93	.66
Recommend the training to others	4.50 (.58)	4.22 (.63)	−1.87	.07	.61

Table 4
Paired sample *t*-test results for perceived competence working interoperably, separated by delivery method.

Item	Mean (standard deviation)				*t*-statistic		Standard error	
	Online		In-person		Online (*p*)	In-person (*p*)	Online	In-person
	Pre	Post	Pre	Post				
Ability to work in multi-agency team	4.21 (.83)	4.43 (.92)	4.03 (.44)	4.08 (.44)	−2.27 (.03)	−1.00 (.32)	.50	.33
Confidence working in multi-agency team	4.21 (.83)	4.43 (.88)	3.97 (.44)	4.19 (.46)	−2.27 (.03)	−3.15 (<.01)	.50	.42
Importance of working in a multi-agency team	4.71 (.81)	4.75 (.80)	4.84 (.37)	4.73 (.45)	−1.00 (.33)	1.43 (.16)	.19	.46
Know what actions to take to facilitate multi-agency working	4.46 (.84)	4.57 (.84)	3.92 (.55)	4.19 (.52)	−1.36 (.18)	−3.65 (<.01)	.42	.45

Table 5
Independent sample *t*-test results comparing change scores between pre- and post-training scores for perceived competence working interoperably, separated by delivery method.

Item	Online M(SD)	In-Person M(SD)	t(df)	*p*
Ability to work in multi-agency team	.21 (.50)	.05 (.33)	−1.48 (44.11)	.15
Confidence working in multi-agency team	.21 (.50)	.22 (.42)	.02 (63)	.99
Importance of working in a multi-agency team	.04 (.19)	−1.08 (.46)	−1.72 (50.58)	.09
Know what actions to take to facilitate multi-agency working	.11 (.42)	.27 (.45)	1.49 (63)	.14

Table 6
Thematic framework for analysis.

Theme	Description	Example quote
Inclusion of psychological theory	Participants' thoughts on the psychological-base for the training tool	"You kind of hit both like opposite ends of the continuum in terms of how you justify what you're talking about, you've given the applied procedural case studies […] the other end of the continuum is that when you talked about academic research" (P2)
Delivery method	Participants' preferred delivery method for training, such as in-person, online, or blended.	"Ultimately, it's always, it's always going to be better if it's presented in person, but if that's not practical, then you just tailor the content to suit" (P2)
Importance of collaboration	Discussions around the importance of training with responders from other organisations.	"I felt quite isolated, it was like, it was easy to get distracted in those moments, if that makes sense with everything else that's going on and I think it'd be helpful to sort of maintain engagement with people by collaborating with the video erm whether that was like a Teams meeting with other people in a room potentially, or if it was in a live room, a physical room, that'd be better" (P6)
Motivation for training	Discussions around the reasons why people complete training and what might prevent them completing training.	"The roadblock that will come up against is the same as for everything else. Unless it's a statutory requirement, or it's some kind of mandatory requirement placed on them, ambulance crews for our service when we run at something like 92–95 % utilisation rates every day, and so trying to get people to leave their day jobs for 20 min to watch a video is a massive challenge, and it's almost like you've got to compel people to do it er and to take crews away for that 20 min, it's 20 min where we haven't got that crew on the road. So there's competing pressures all the time" (P4)

participants could engage with the material.

3.2.3. Importance of collaboration

Several participants highlighted that one of the main advantages of in-person training over online delivery is the opportunity for networking and collaboration with responders from other organisations. One participant expressed feeling "quite isolated" during the online training they received during the study and believed that engaging with others in a physical or virtual room would have improved their experience and maintained their attention longer.

Participants criticised online training for missing out on key opportunities to communicate with responders from other organisations and learn from their experiences. They felt that considering the context of the training through the perspectives of others in the room, not just their own experiences, was crucial. As one participant noted, "when we're talking about like different agencies working [...] with each other, I think it really helps when you're able to speak to people that are a similar level to you like peers, people you're likely to come into contact with operationally as well [...] so you're [...] able to collaborate. [During online training] you miss out on a lot of those [...] incidental conversations and sharing of experiences, and what that allows to happen is for [...] that arrogant perspective of what we do is right [...] it stops that happening because it allows people to break down their experiences and consider [...] their own experiences and opinions against the context of the people sat next to them" (P6).

Additionally, participants found that receiving training alongside responders from other organisations allowed them to practice the collaboration techniques being taught. One participant mentioned, "If you've got a room full of police, fire, and ambulance, then having someone deliver it [...] you could [...] practise what you preach in terms of making that interoperability and making people sort of share learning and talk to each other" (P7).

3.2.4. Motivation for completing training

Nearly half of the participants discussed their motivations for completing the training. Some felt that online training packages are not considered important among their peers and are often completed just to check a box, rather than for genuine learning. They argued that making training mandatory is not the most effective way to motivate people to complete the training. One participant noted, "a lot of people don't pay the learning packages [...] the importance that they actually are, they think [...] because it's just mandatory [...] they get it out of the way for a couple of years [...] rather than wanting to learn from it" (P1).

However, other participants believed that due to competing pressures in the emergency services, making the training mandatory is essential for ensuring its completion. For example, one participant said, "we've got lots of competing pressures in different services, particularly ambulance and fire sectors. Having something quick and accessible [...] because commanders all have to maintain their competency in currency through [continuing professional development] activities" (P3).

4. Discussion

This paper evaluated a novel psychology-based training programme developed using the Social Identity Approach to enhance interoperability among emergency responders. We aimed to understand whether participants found the psychological elements of the training useful in improving their knowledge about how to facilitate a more effective multi-agency response. We also explored participants' engagement with the training and their perceived competence working interoperably, specifically comparing online and in-person delivery methods.

Overall, participants responded very positively to the training. They particularly valued the inclusion of psychological insights that explained why particular actions are important for effective interoperability. This helped them understand how these behaviours support better collaboration. Many participants said that they appreciated this aspect because it was new to them and gave them context behind the science that underpins the JESIP principles.

The findings of this evaluation support the growing recognition that enhancing interoperability among emergency responders requires not just structural or procedural coordination, but also a shared psychological understanding among responders [18,37]. This also aligns with research in medicine, where the development of shared mental models has been shown to improve team performance, communication, and decision-making in high-pressure environments [55,56]. Shared mental models refer to the common understanding among team members regarding tasks, roles, and team dynamics, which enable coordinated and adaptive responses under stress. These concepts complement the Social Identity Approach, which emphasises the importance of a shared sense of "us" and psychological group membership as the basis for effective collaboration [22]. While shared mental models focus on cognitive alignment, the Social Identity Approach adds a motivational layer related to the capacity for shared social identity to foster trust, mutual support, and commitment to joint goals. Together, these frameworks offer a powerful lens for understanding how interoperability among responders is achieved through alignment that is both cognitive and social. Future research should consider how these approaches can be integrated to maximise their impact on multi-agency collaboration.

Nearly all participants, regardless of whether they received the training online or in-person, found the training easy to follow and understand. Most also felt the training was important and said they would recommend it to others. Whilst interview participants tended to prefer in-person delivery, survey results showed no difference in satisfaction between the two delivery methods. This contrasts with previous research, which found that online training is usually rated less favourably than in-person training [35]. One possible explanation for the satisfaction across both delivery methods in this study is the widespread awareness of the ongoing problems with interoperability [3] and the attention it has received in recent incident inquiries [4,5]. Given this context, responders may view any training that addresses interoperability, regardless of how it is delivered, as valuable and necessary.

Perceived competence for working interoperably increased after both versions of the training. Notably, participants who completed the online training reported feeling more confident in their ability to work within a multi-agency team, but their understanding of the specific actions required to deliver an effective multi-agency response did not improve. In contrast, those who attended the in-person training showed an increase in their knowledge of actions that enhance collaboration. This finding differs from previous research, which found no difference in knowledge gain between online and in-person training [35]. A possible explanation for this difference is the collaborative nature of the training content. The in-person delivery allowed participants to interact directly with responders from other organisations, enabling them to discuss and explore practical collaborative strategies in real time. This likely facilitated greater learning than in the online delivery. Supporting this, interview participants emphasised that in-person training provided valuable opportunities to engage with responders from different organisations. They noted that face-to-face settings allowed them to practice the collaboration techniques introduced in the training, which helped reinforce key lessons and translate theory into action.

In the online delivery, most participants felt the training would be more beneficial for improving others' ability to deliver effective multi-agency responses than their own. This may be because nearly all of them were already 'very' or 'extremely' familiar with JESIP before the evaluation, suggesting a strong existing knowledge of interoperability. In contrast, for participants in the in-person delivery there was a smaller gap between how useful they thought the training was for themselves compared to others. The reasons for this difference are unclear, but as noted earlier, the in-person format encouraged more in-depth discussions among participants. These conversations may have led to greater reflection on both their own skills and those of others.

Interview participants emphasised that it is not just the method of training delivery that is important, but also the reasons why people choose to take part. Many discussions focussed on mandatory training and whether it effectively motivates participation, although there was no clear agreement on its impact. Recent research shows that many emergency responders often complete training in their own time, outside of work hours [57]. At the same time, limited operational capacity has been identified as a key barrier to delivering necessary training ([36]; cf. [27,28]). This suggests that while mandatory training may help by creating a space for learning during work hours, its effect on motivation remains unclear. Future research should explore what drives participation in interoperability training and how factors such as mandatory requirements and operational capacity influence responders' engagement and motivation to complete training.

In the current study, participants completed the training before responding to questions about their engagement with it and their perceived competence in working interoperably. This approach provides valuable insights into both the effectiveness of the training and areas for development. However, future research would benefit from evaluating the training in more realistic environments that closely mimic the complexities of real incidents, and which therefore encourages behaviour that more accurately reflects real-world responses [58]. Such environments could include field exercises (e.g., Ref. [59,60]) or immersive simulations (e.g., Ref. [61]). To effectively assess the impact of training on behaviour, future evaluations should also consider incorporating these types of exercises into their methodology.

Culture is a crucial factor in efforts to change behaviour, yet it has often been overlooked in attempts to improve interoperability previously [3,37]. Power and colleagues [37] argue that one reason JESIP has struggled to enhance interoperability is the failure to embed it within the organisational culture of the emergency services. They highlight three psychological principles as essential for fostering an interoperable culture: establishing trust, building secure team identities, and setting cohesive goals. Previous research demonstrates that these psychological principles are interrelated rather than discrete, showing that establishing a shared identity can promote trust and support discussions around shared goals [18]. This study therefore makes an important contribution by developing and evaluating training aimed at fostering a shared social identity within emergency response teams – a vital step towards creating an interoperable culture.

The integration of theory-into-action is critical in applied research, particularly in fields where training and preparedness have real-world implications, such as in emergency response. Theory-into-action research bridges the gap between conceptual understanding and practical implementation, ensuring that theoretical frameworks are tested, refined, and made relevant through real-world application. This approach helps to address the well-documented research-practice gap within the field of emergency management [,25,26]. Promoting such research enhances the utility of academic work, making it more accessible and impactful for practitioners and policy-makers. The current study contributes to this effort by exploring how a novel psychology-based training programme might change responders' ability to work in a multi-agency team. The results from this can inform the ways in which training is tailored and delivered to emergency responders, promoting knowledge transfer and skill development. Nevertheless, future research should build on this study and continue to evaluate training programmes not only for their theoretical grounding but also for their capacity to generate actionable insights that inform practice and policy.

4.1. Strengths and limitations

This paper integrates social psychology into interoperability training based on recent research which highlights the important role that psychology plays in the dynamics of interoperability [15–18]. Our evaluation of a novel training programme informed by this previous empirical work advances efforts to address challenges faced by emergency responders during multi-agency responses. However, there are limitations which need attention.

First, the in-person sample consisted entirely of Ambulance responders, which may have skewed the results. Additionally, participants were not randomly assigned to delivery methods (in-person vs. online), and this may have introduced bias into the study. However, the online sample included a near-equal mix of Police, FRS, and Ambulance responders, and there were no significant differences between services in their responses to key measures. This suggests that the views of Ambulance responders are similar to those of other services, although further research is needed to confirm this.

Another limitation is the focus solely on the blue-light services, excluding other Category 1 responders such as local authorities and NHS bodies. This reflects a broader gap in interoperability research, and future research should aim to include a more representative range of response organisations that would typically be involved in major incident response.

Most participants also reported being familiar with JESIP, which may limit how well the findings apply to those who have less exposure to JESIP. This familiarity might explain why there was no significant change in the way that participants rated the "importance of working in a multi-agency team" after the training, because many were already aware of interoperability and accepted the value of JESIP principles. To address this issue, future research should include responders with varying levels of knowledge and experience with interoperability to better assess the impact of training.

Although all 65 responders who took part in the evaluation were invited to take part in a follow-up interview, only seven were able to commit to this phase of the study. This low follow-up reflects a common challenge in emergency response research associated with the difficulty with engaging hard-to-reach populations who have ongoing operational commitments. Limited availability, shift patterns, and the often-unpredictable nature of their roles constrain opportunities for in-depth follow-up. Whilst the insights gained from the interviews remain valuable for the current evaluation, future research should explore other methods of engagement to enhance participation and ensure broader representation in similar evaluations.

Finally, while psychological insights are important, they alone cannot solve interoperability complexities. Other factors, such as technical communication systems like those used in the Manchester Arena Attack [4] and the London bombings [62] are also important. Thus, while psychology-based training is beneficial, other factors must also be considered to enhance interoperability.

4.2. Conclusion

This paper evaluated a psychology-based training programme developed using the Social Identity Approach to enhance interoperability among emergency responders, comparing online and in-person delivery methods. Overall, participants liked the training and appreciated the inclusion of psychological principles and understanding of how these could improve interoperability. Despite a preference for in-person training in interviews, survey data showed no difference in satisfaction between delivery methods. Both delivery methods increased participants' confidence in multi-agency teamwork, but in-person training enhanced knowledge of specific collaborative actions more effectively. The need for accessible, jargon-free content was highlighted to avoid barriers to understanding. While mandatory training could increase operational capacity, its impact on motivation remains unclear. Future research should explore this area further to optimise training effectiveness for emergency responders.

5. Practitioner points

- Training programmes which focus on improving interoperability should include psychological principles, such as social identity processes, to help emergency responders understand the reasons behind certain actions and how these actions can enhance interoperability.
- Interoperability training content should be presented in a jargon-free and accessible format. This will help prevent any barriers to understanding and make the training more accessible for all participants, regardless of their academic background.
- While both online and in-person delivery methods for interoperability training can increase participants' confidence in multi-agency teamwork, in-person training can enhance knowledge of specific collaborative actions more effectively. Training providers should therefore consider the benefits of both methods and aim to balance them, possibly integrating elements of each to optimise interoperability training outcomes.

CRediT authorship contribution statement

Louise Davidson: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Holly Carter:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **John Drury:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Richard Amlôt:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **S. Alexander Haslam:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Ethical considerations

Ethical approval was not sought for the present study because it was classed as a service development.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Louise Davidson reports financial support was provided by Fire Service Research and Training Trust and the University of Sussex. Holly Carter, John Drury reports financial support was provided by Economic and Social Research Council. Holly Carter, Richard Amlot reports financial support was provided by National Institute for Health and Care Research.

Acknowledgements

This evaluation was carried out as part of a PhD project by Louise Davidson at the University of Sussex in collaboration with the UK Health Security Agency (UKHSA). The PhD is jointly funded by the Fire Service Research and Training Trust and the University of Sussex. John Drury and Holly Carter were supported by a grant from ESRC reference ES/V005383/1. Holly Carter and Richard Amlôt are funded by the National Institute for Health Research Health Protection Unit (NIHR HPRU) in Emergency Preparedness and Response (EPR), a partnership between UKHSA, King's College London and the University of East Anglia and the NIHR HPRU in Behavioural Science and Evaluation in partnership with the University of Bristol. Louise Davidson is also affiliated to the EPR HPRU. The views expressed are those of the authors and not necessarily those of the NHS, the NIHR, the Department of Health and Social Care or UKHSA. For the purpose of open access, the author has applied a Creative Commons Attribution (CC BY 4.0) licence to any Author Accepted Manuscript version arising.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijdrr.2025.105725.

Appendix 1

Initial feedback from responders, reported as per the GRIPP2 reporting framework (Staniszewska et al., 2017)

Section and topic	Item
1. Aim	To collaboratively engage responders in the initial development stage of the training.
2. Methods	The content of the training was sent in a basic PowerPoint format to eight responders from the Police (*N* = 2), FRS (*N* = 3), and Ambulance Service (*N* = 3). Responders provided either written feedback (N = 2) or face-to-face to the lead researcher (*N* = 6). Responders were specifically asked what they liked and did not like about the content of the training, what areas could be improved, whether they understood what was meant by 'shared identity', and whether there were any areas that they did not understand.
3. Results	Responders who reviewed the training content said they liked it, and it made a valuable contribution to understanding interoperability. However, several suggestions were also made to improve the training, including: more detail on where challenges arose during the Manchester Arena response and the provision of specific examples; emphasising the importance of building relationships before an incident; and making the information presented on the slides more concise. Several responders liked the psychology content and advised on the inclusion of more detail of the psychology elements to make the contribution of psychology clearer. Finally, most responders advised to run the evaluation of the training with responders across all levels of command (operational, tactical, and strategic) because the elements included in the training are applicable across all levels.
4. Discussion	The content of the training was updated based on the feedback from responders. Specific examples where the training was updated include examples from the Manchester Arena inquiry of the delayed attendance from the FRS, adding additional slides expanding on the psychological elements of *how* a shared identity can be used to improve group working, and reducing the amount of information on the slides by breaking down sentences into short bullet points. In addition, formatting was added to the presentation, such as animations, transitions, and colour. A similar colour scheme to that used in JESIP documents and training (red, green, and blue) was used to allow for consistency with other training that responders are familiar with. The lead researcher recorded themself presenting the training, and the PowerPoint was turned into video format. The video was 29 min, 27 s in length. It was uploaded onto YouTube as an unlisted video: https://www.youtube.com/watch?v=uubrBT2rYWc.
5. Reflection	The feedback received from responders was embedded as far as possible into the training tool. Due to the near equal engagement across the emergency services, feedback from each service was able to be represented. Whilst written feedback was less comprehensive than the verbal feedback, taken together with the verbal feedback, several areas for change and improvement were able to be identified and implemented into the finalised training.

Appendix 2

Post-hoc sensitivity analysis

The post-hoc sensitivity analysis using G*Power [53] indicated that while some of the statistical tests had sufficient power, others did not fully meet the required threshold for detecting the expected effect sizes. Specifically, for the paired sample *t*-test with 65 participants, the effect sizes ($ds = .42, .45, .37, .44$) exceeded the sensitivity threshold of $d = .35$, indicating sufficient power. Similarly, the one sample *t*-test with 65 participants yielded effect sizes ($ds = .70, .71, .72, .71, .58, .65, .61$) above the threshold of $d = .35$.

However, the independent sample *t*-test with 28 participants in one group and 37 in the other showed effect sizes ($ds = .70, .71, .72, .71, .59, .66, .61$) that partially fell below the sensitivity threshold of $d = .71$, indicating the test was partially underpowered. Additionally, the paired sample *t*-test with 28 participants revealed effect sizes ($ds = .50, .50, .19, .42$) below the threshold of $d = .55$ and the paired sample *t*-test with 37 participants had effect sizes ($ds = .33, .42, .46, .45$) below the threshold of $d = .47$. These results suggest that these tests were underpowered and might not detect smaller effects, potentially limiting the generalisability and robustness of the findings.

Data availability

Data will be made available on request.

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